

Effective Halo and Starfield Measurements on HDR Displays

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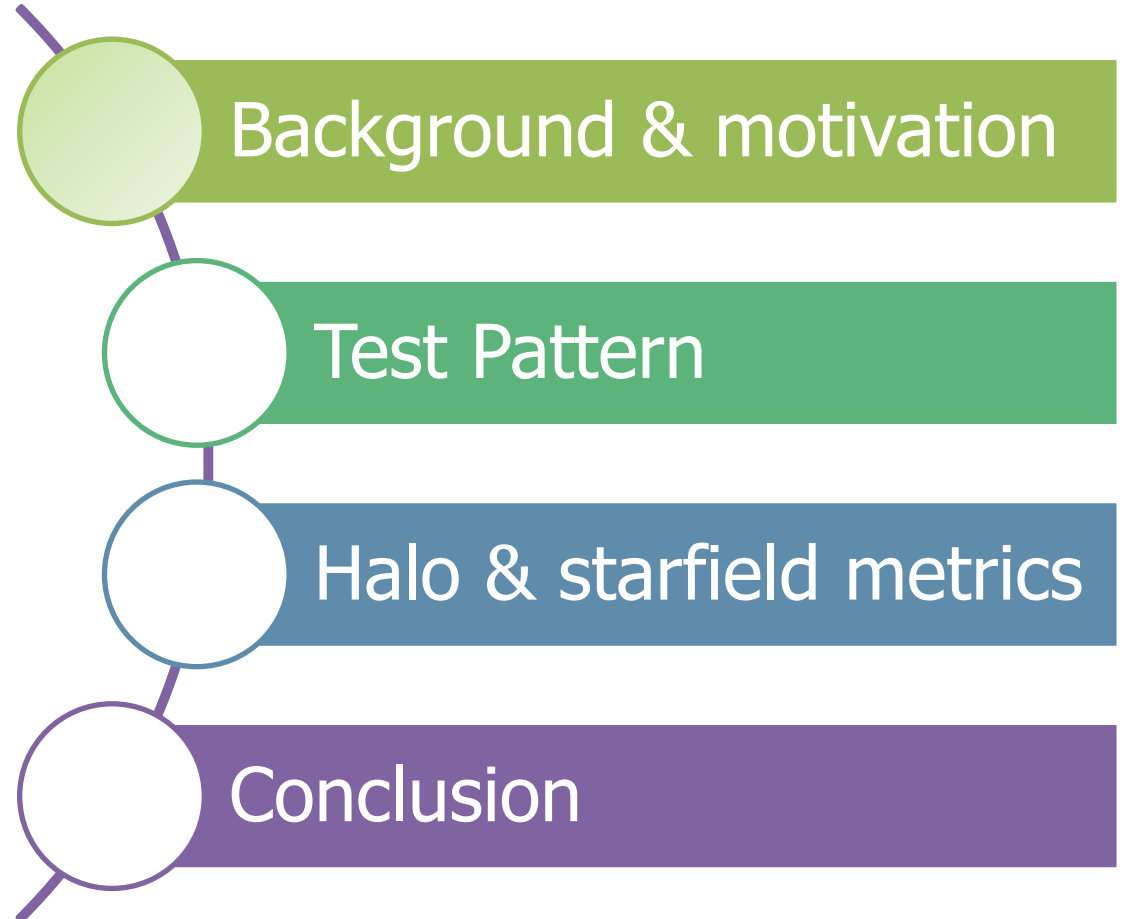
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Outline

Effective Halo and Starfield Measurements on HDR Displays



Background

Halo | Halation | Blooming

800,000:1 [1% area white 400cd/m^2 , Full black 0.0005cd/m^2]



Background

Starfield



Background

Starfield

actual

intent



Background

IDMS method - Halation

5.23 Halation

DESCRIPTION

We measure the luminance of a black box with a white background as the size of the box is adjusted from a small fraction of the screen to full screen. Halation is said to occur when light from surrounding white areas corrupts a black area on the screen. This measurement is a method to characterize the amount of halation for a black box at the center of a white screen.

SETUP

As defined by these icons, standard setup details apply (3.2).

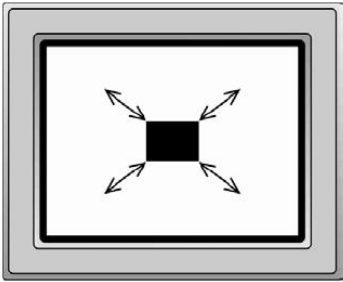


Figure 5.23.1. Expanding black box on white background for halation measurement.

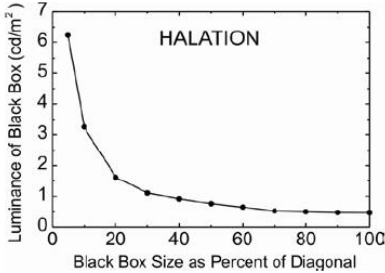


Figure 5.23.2. Example of halation data.

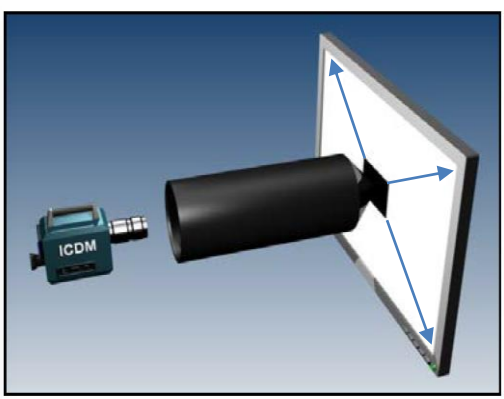
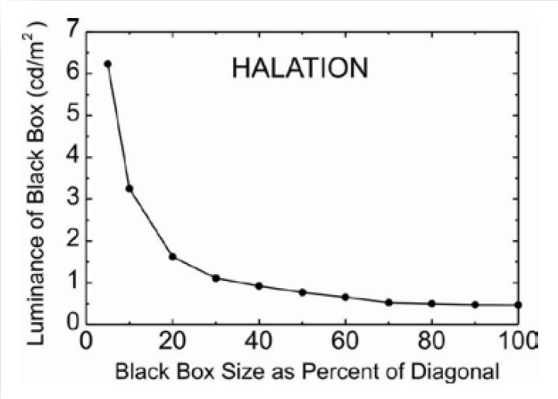


Table 5.23.2. Analysis example.

Sample data only: Do not use any values shown to represent expected results of your measurements.

Box % Diag.	L_{box} (cd m ⁻²)
5 % (L_{max})	6.23
10 %	3.25
20 %	1.62
30 %	1.11
40 %	0.923
50 %	0.769
60 %	0.655
70 %	0.523
80 %	0.498
90 %	0.473
100 % (L_{B})	0.468
Halation	6.0 %

Background

IDMS method – Starfield Contrast

5.12 Starfield Contrast

DESCRIPTION

We determine the contrast for a display with content-dependent control (such as dynamic contrast, global dimming, local dimming, etc.) at the threshold of activation of the dynamic control by use of a set of controlled ICDM-provided test patterns to help establish a realistic low level for black content and determining the contrast at the threshold. Figure 5.12.1 shows a typical measurement setup for a star pattern with a center black box; the starfield must be masked off to prevent veiling glare in the detector especially when measuring the black box. The figure shows the use of a SLET (stray-light-elimination tube, see A2.1.5 in the appendix for more information) as an example of several ways to mask the area to be measured.

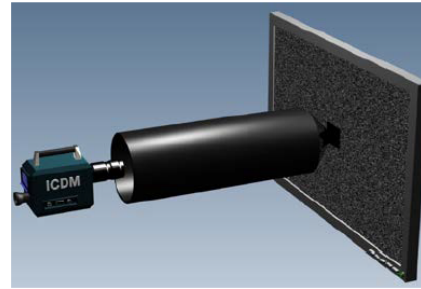
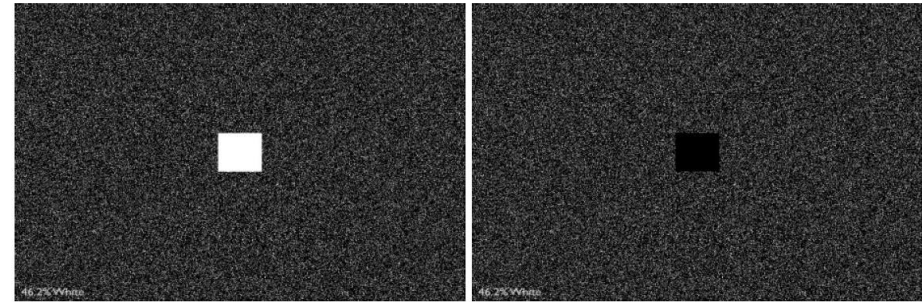


FIGURE 5.12.1. Starfield contrast measurement setup. A mask must be used for all black measurements so light from the surrounding imagery will not produce veiling glare in the LDM.

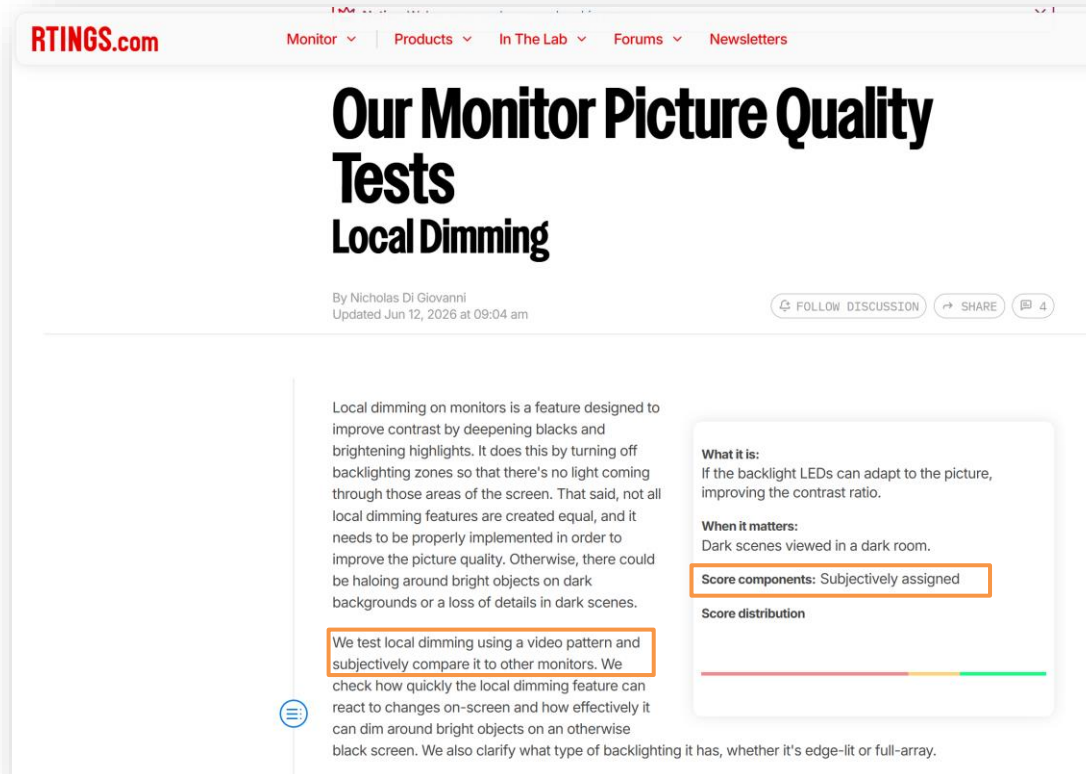


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Starfield doesn't measure the stars luminance between different steps. It only measures the contrast (White/Black).
No information on the stars clipping effect.

Background

Subjective Evaluation



<https://www.rtings.com/monitor/tests/picture-quality/local-dimming>

Motivation

Effective method to test and quantify

❑ Issue on some of the higher tiers HDR displays

- Especially for FALD

❑ Dimming zones and algorithm

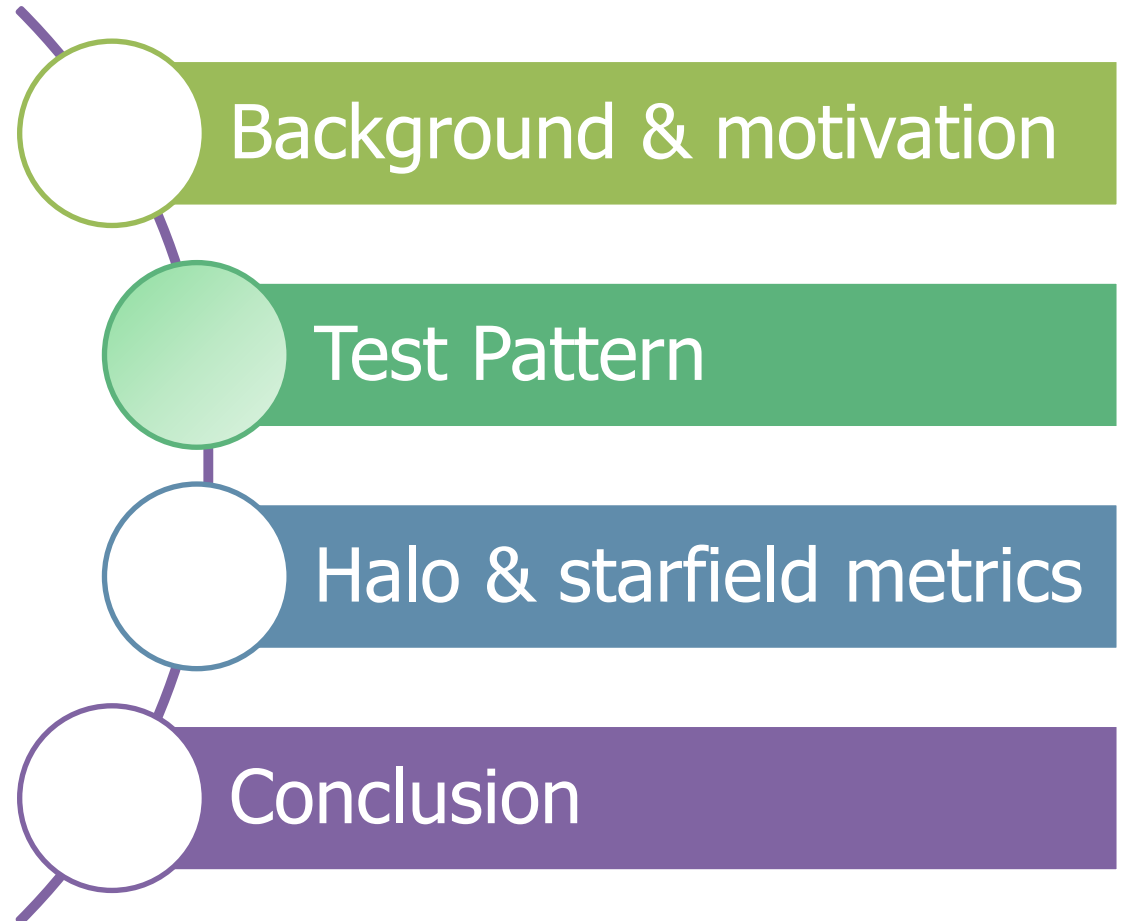
- Halo vs Starfield trade-off - improving one by sacrificing the other

❑ Need better solution

- Common test pattern – leverage existing when possible
- Objective metric
- Works across different size, resolution, pixel density, ...
 - Technology independent – across different backlight configuration and algorithm

Outline

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Test Setup

❑ Spot LMD (light measuring device)

- Compatible with contact / non-contact
- 27mm measurement diameter is critical to avoid measurement error in Starfield

❑ Frustum mask or flat mask

- Eliminate veiling glare and stray light

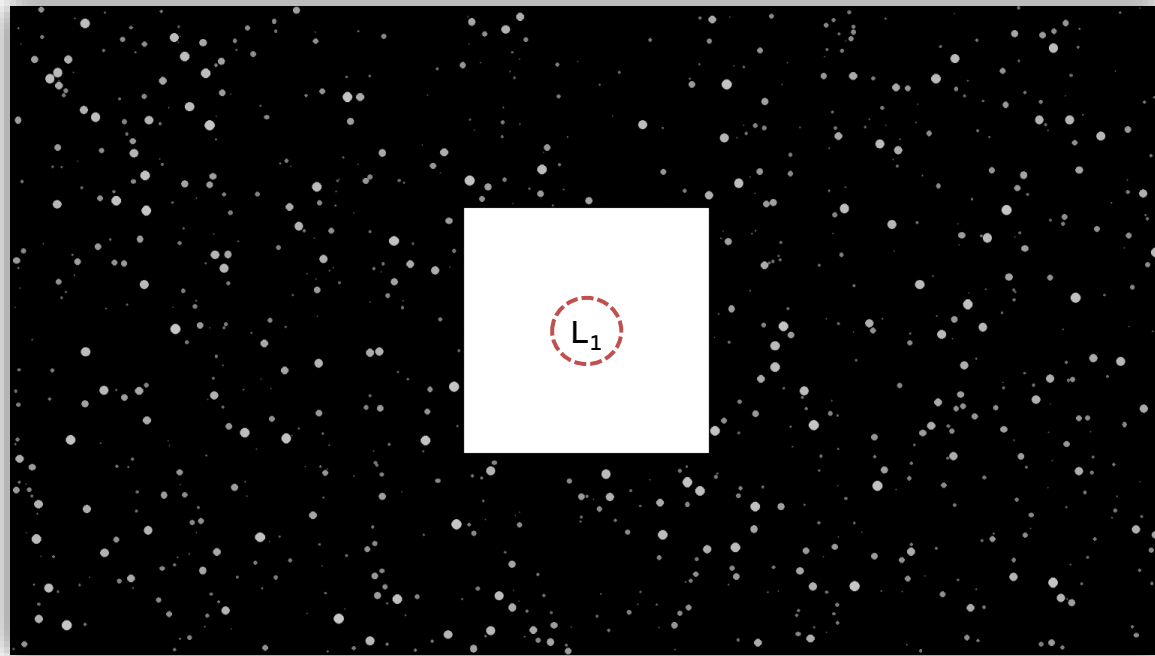
❑ Test Pattern

- Derive from VESA DisplayHDR Test software

Test Pattern

Reference Luminance

□ 8% Center Patch with 2% APL background

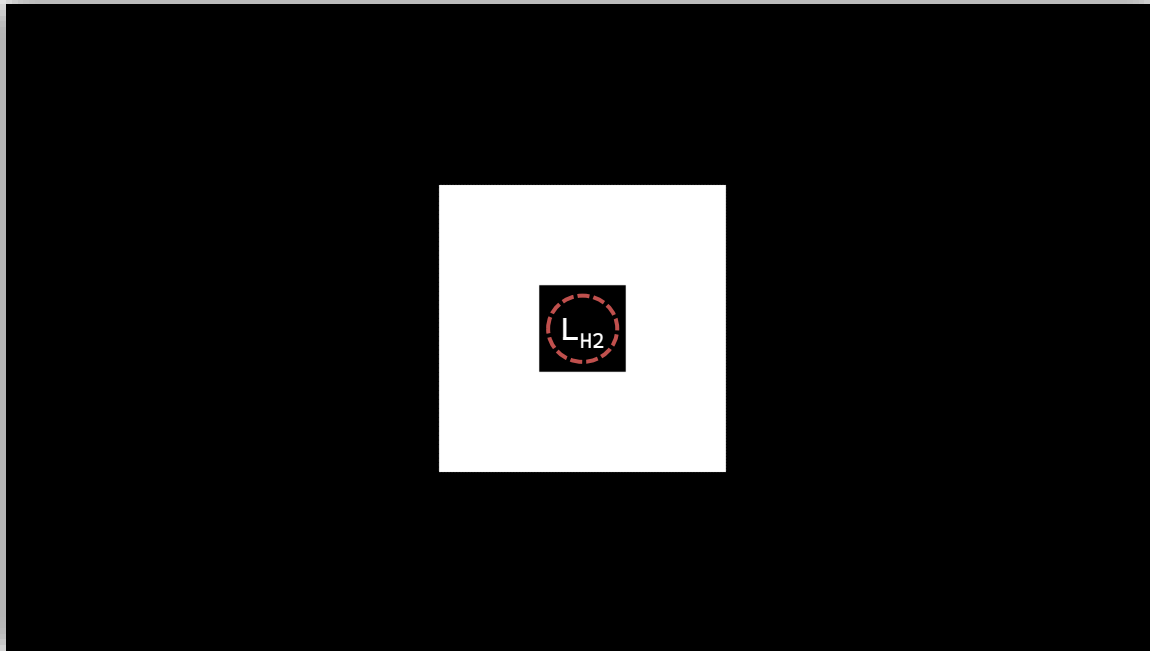


Test Pattern

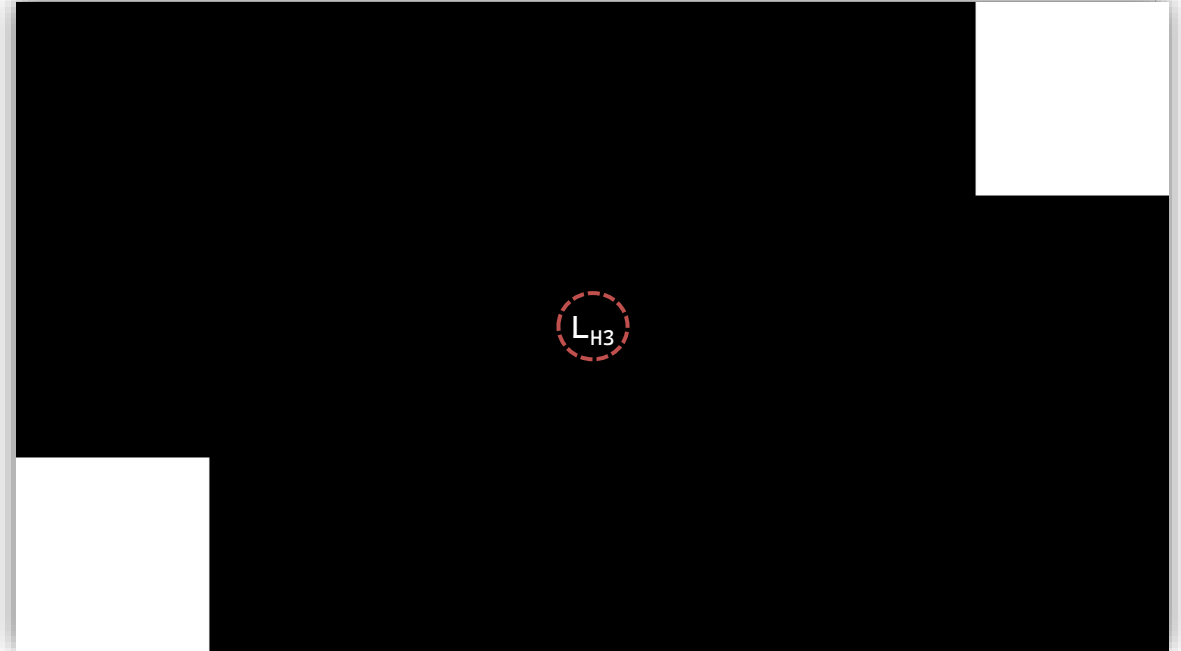
Halo

□ 10% Center square patch

- Fixed 40mm x 40mm center (Black)



□ 5% Dual Corners

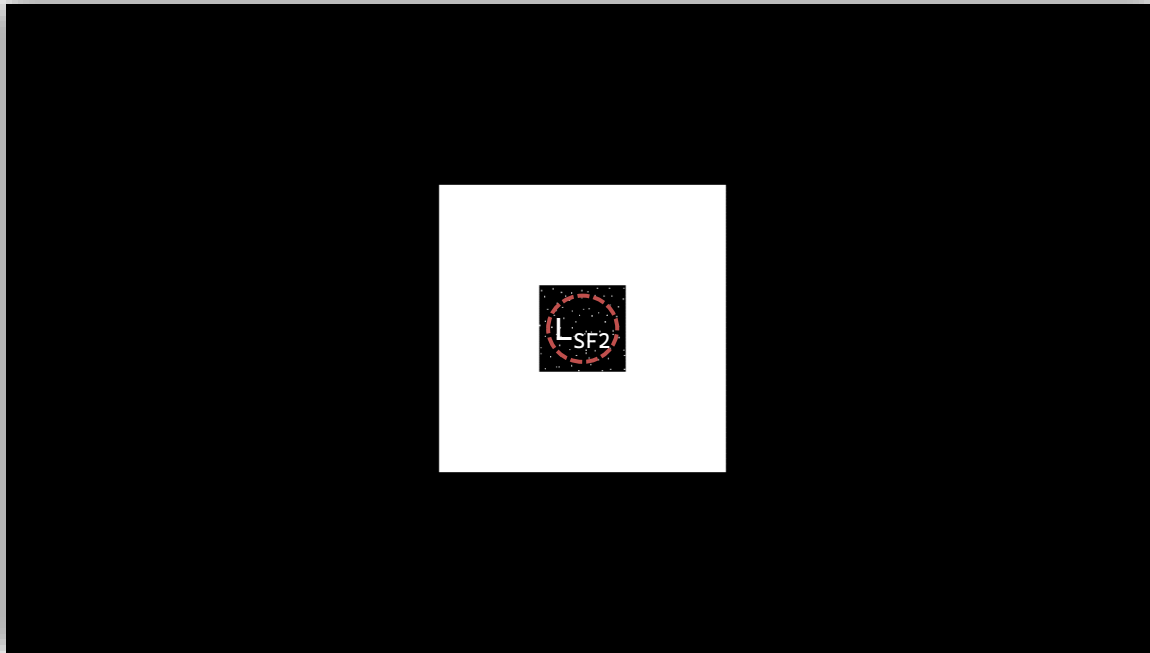


Test Pattern

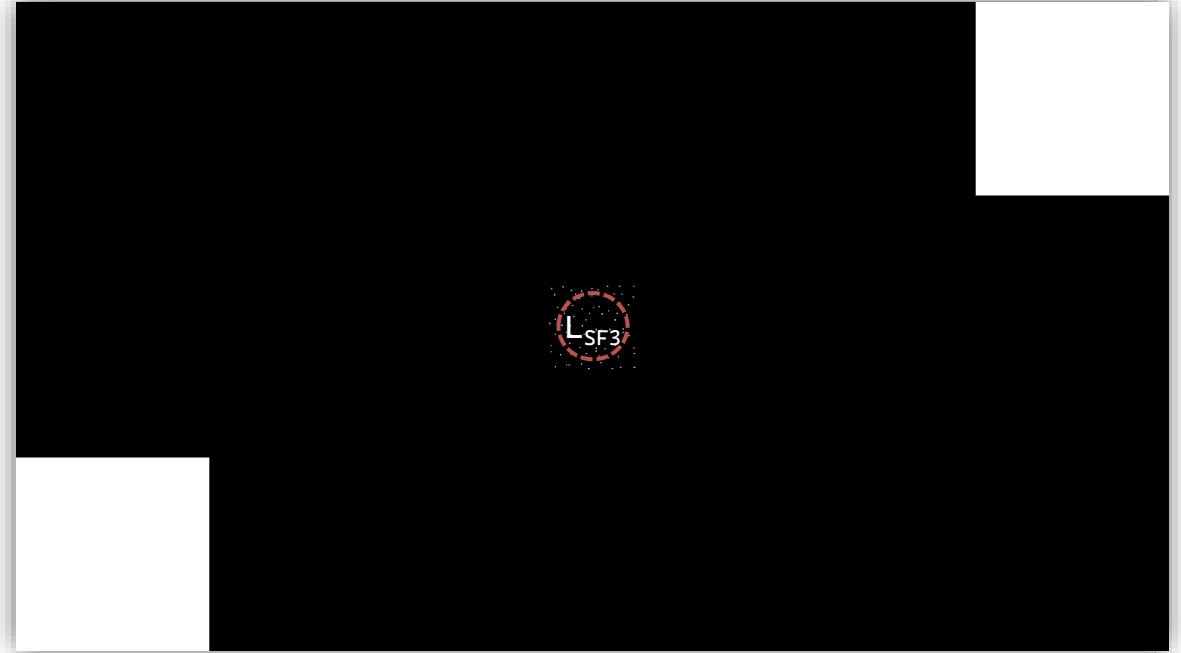
Starfield

□ 10% Center square patch

- Fixed 40mm x 40mm center with fixed star density



□ 5% Dual Corners

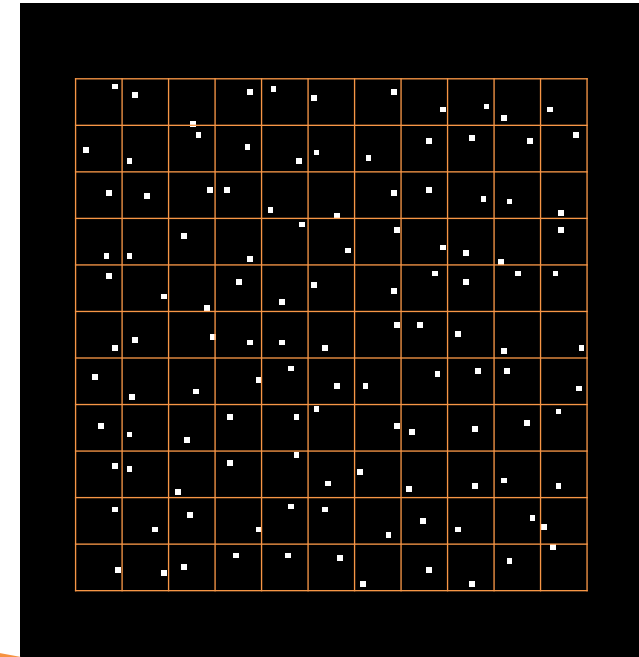
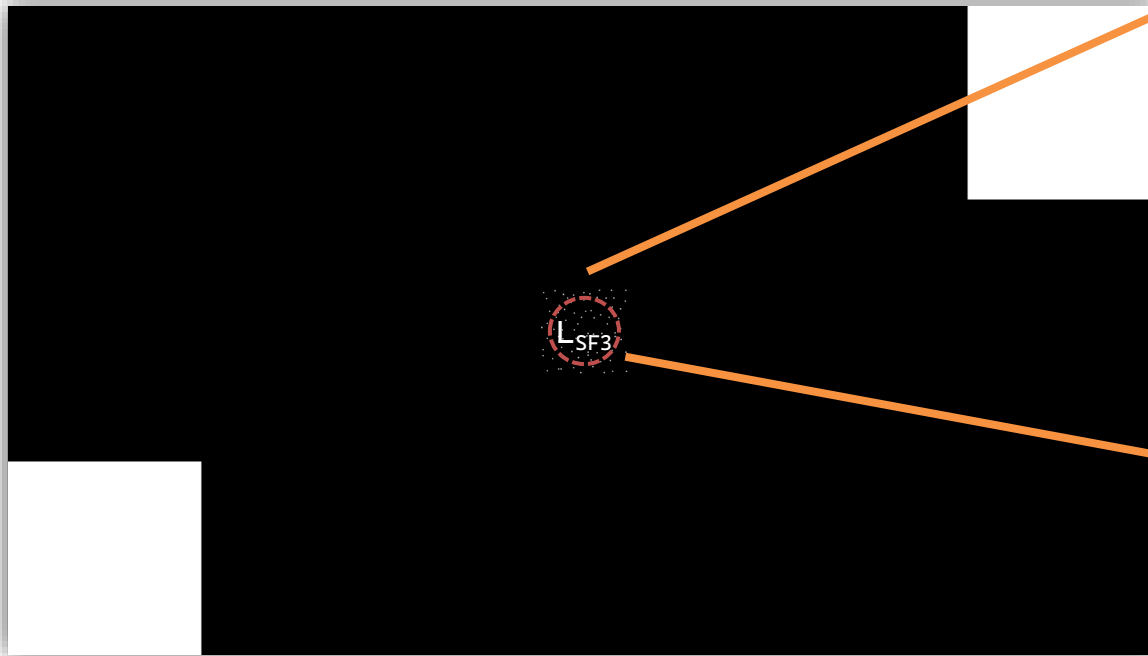


Test Pattern

Starfield Density Distribution

□ 1/64 pixel density

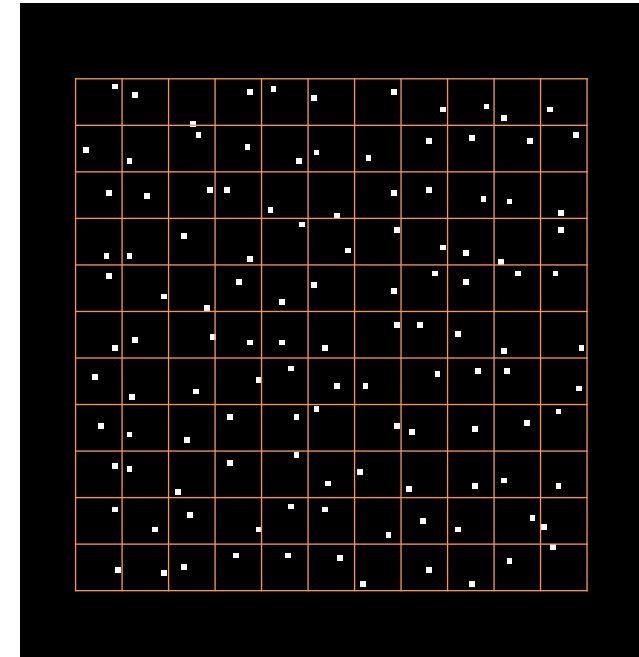
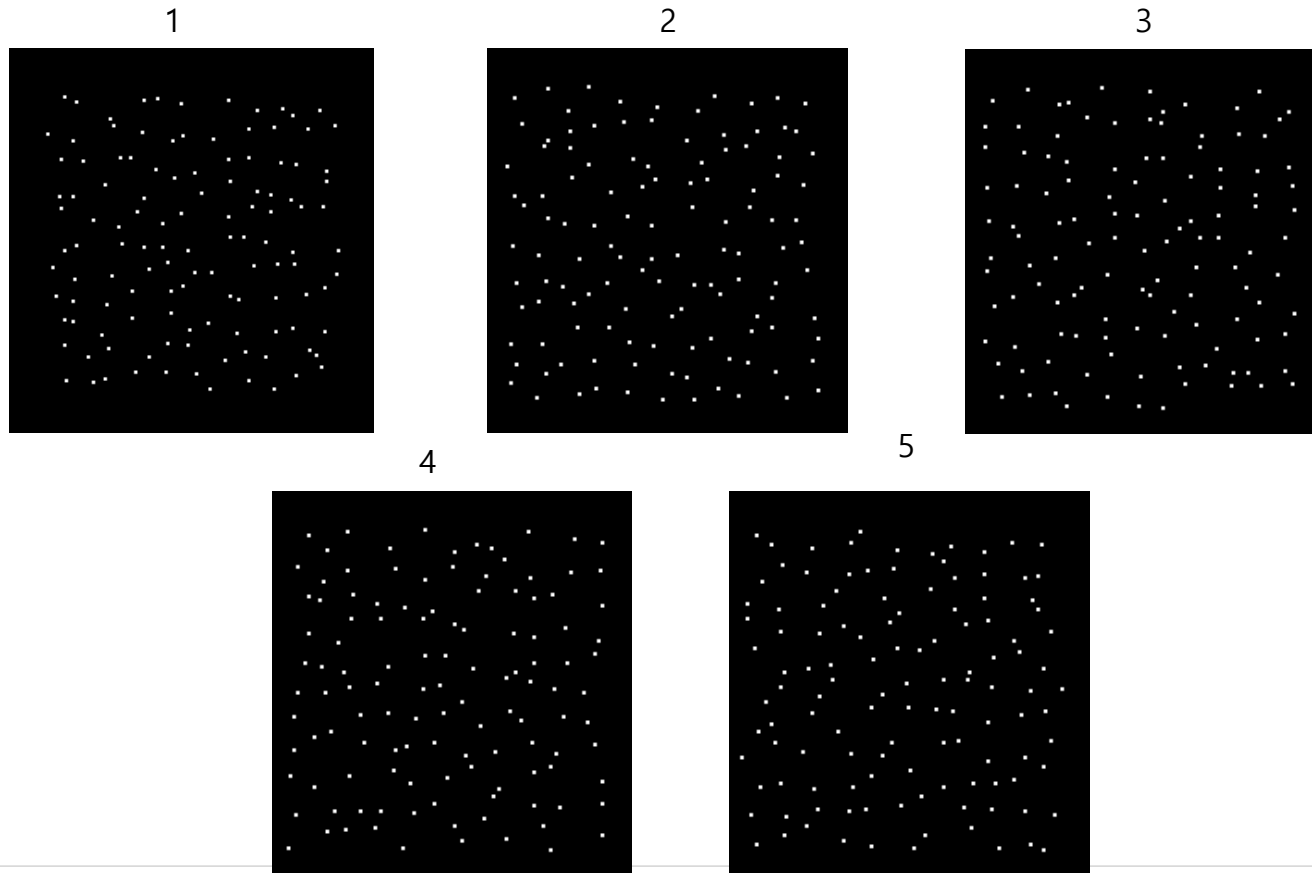
- 40 x 40mm split into grid, with distribution of a 4 star pixels / 256 pixel.



Test Pattern

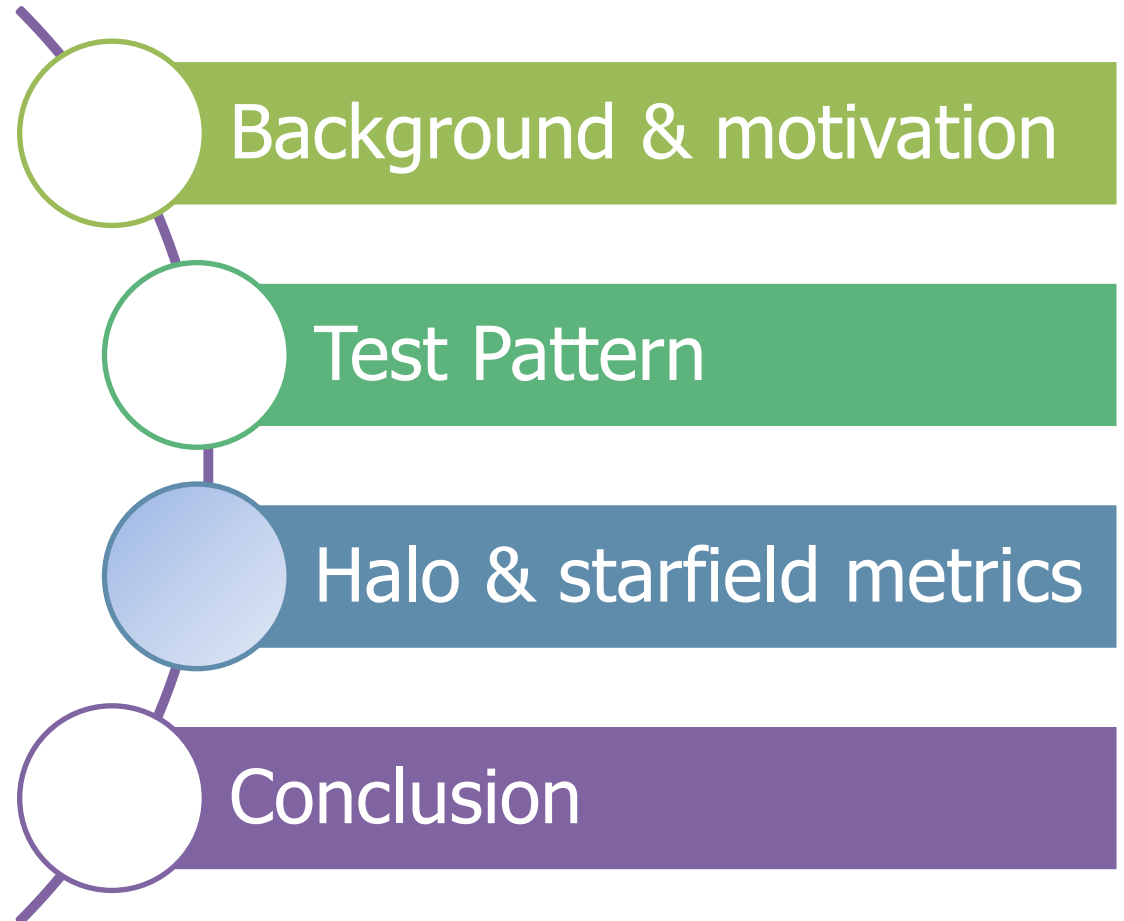
Starfield Density Distribution

❑ Multiple patterns based on different random seed



Outline

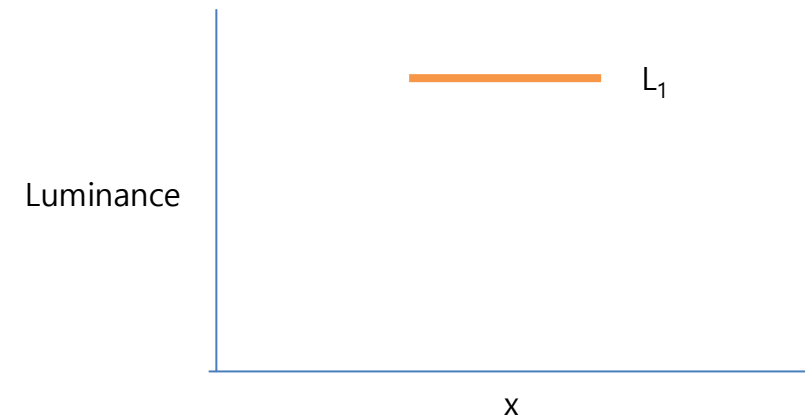
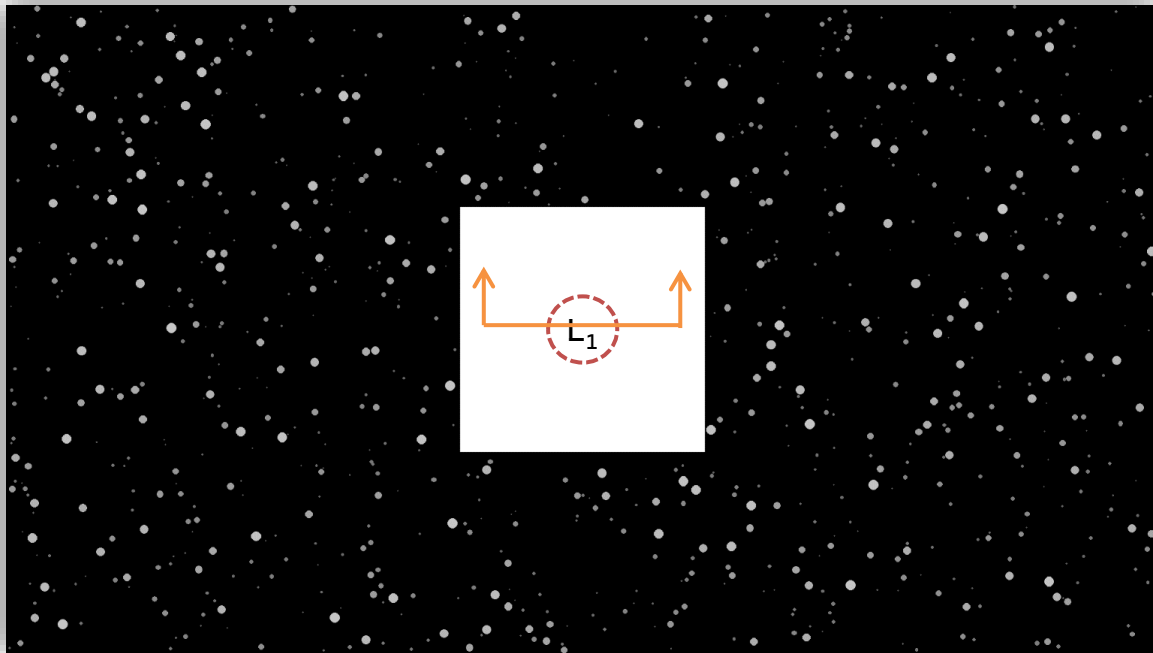
Effective Halo and Starfield Measurements on HDR Displays



Measurement

□ intersection view

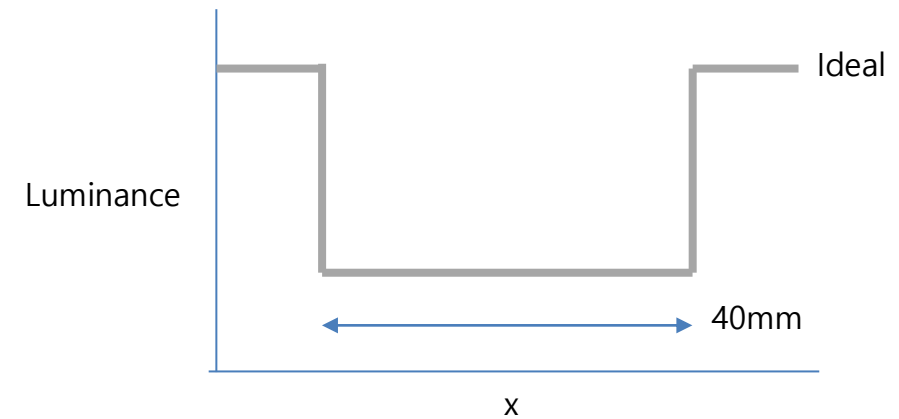
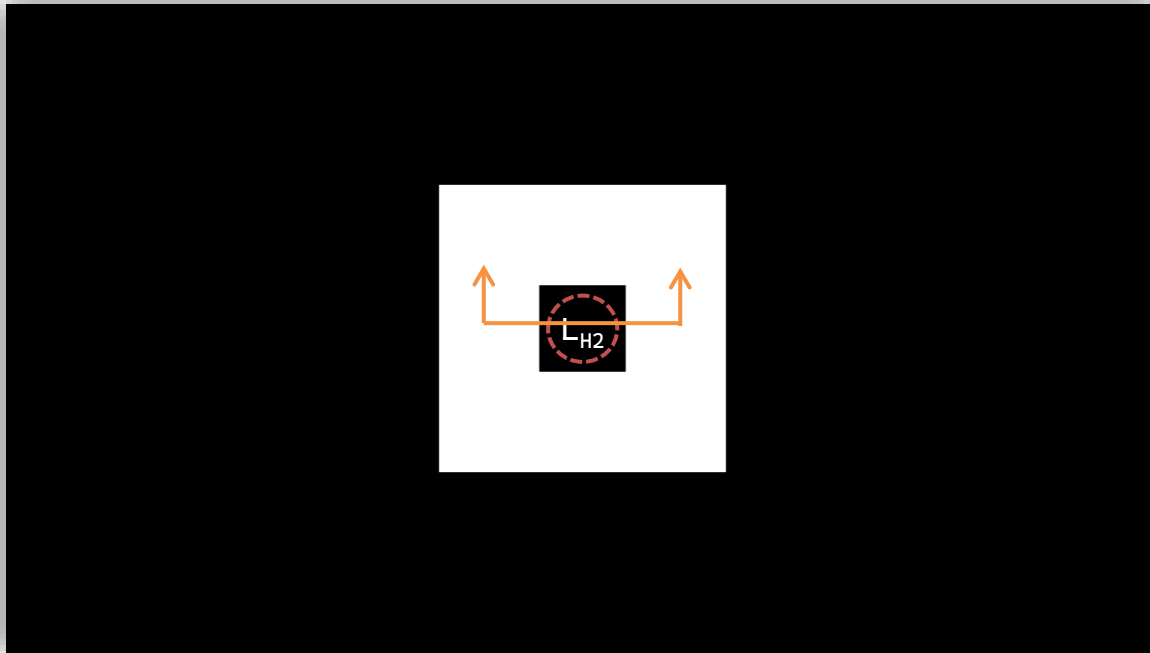
- Distance vs luminance



Measurement

□ intersection view

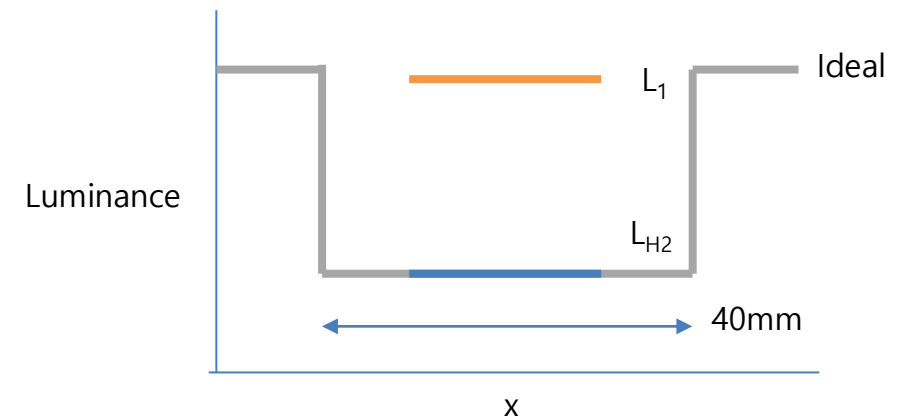
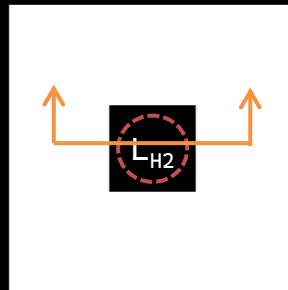
- Distance vs luminance



Measurement

□ intersection view

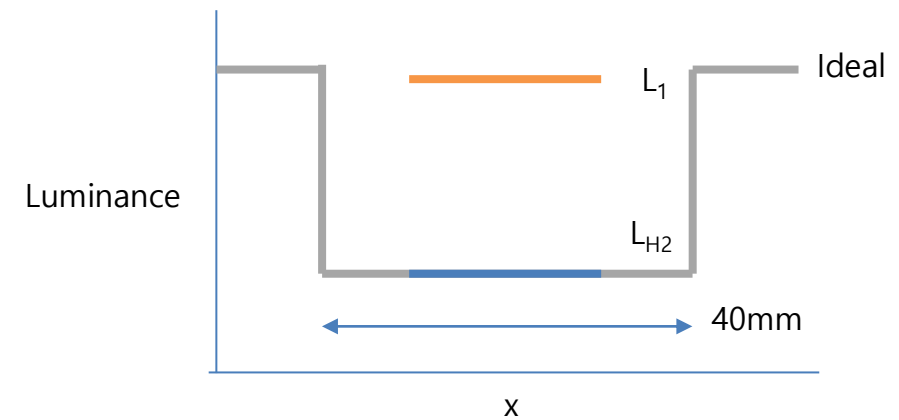
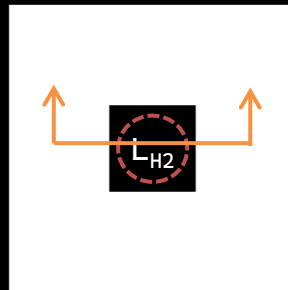
- Distance vs luminance



Measurement

□ intersection view

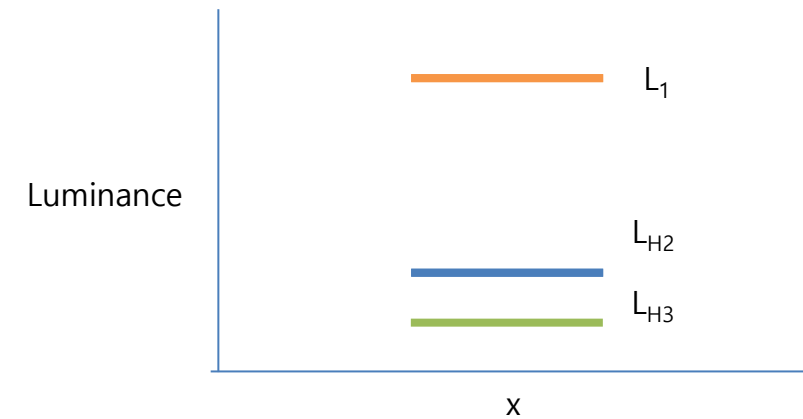
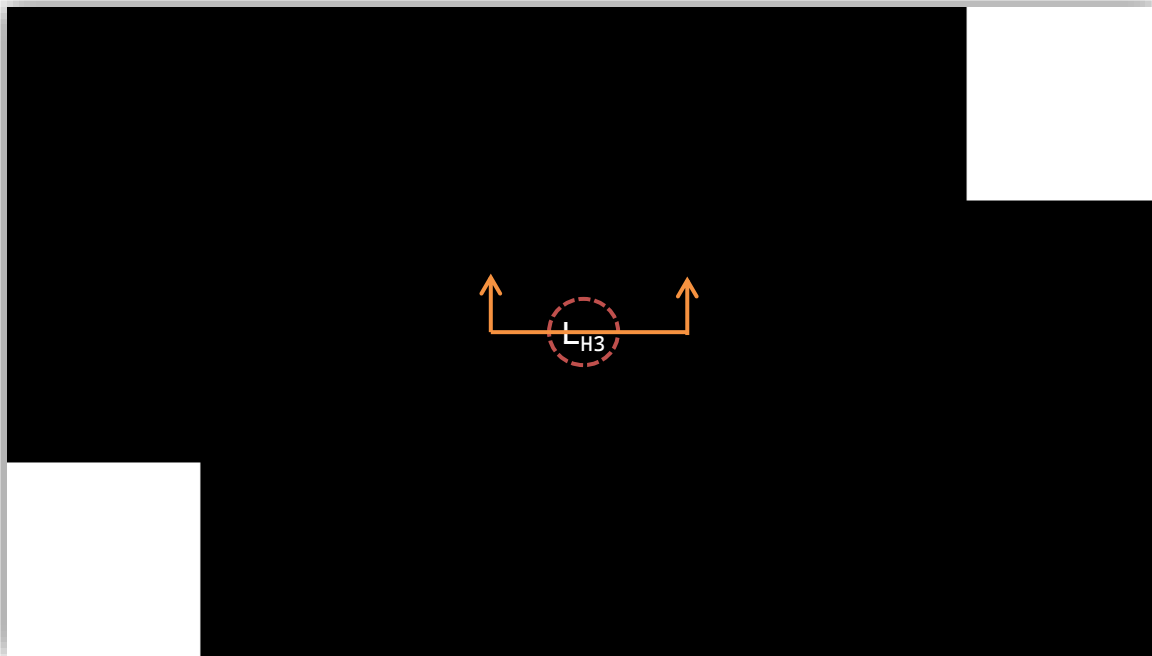
- Distance vs luminance



Measurement

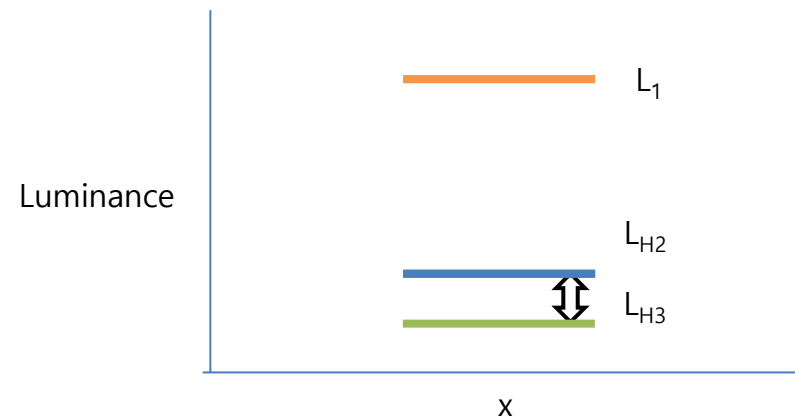
□ intersection view

- Distance vs luminance



□ Result in # of stops

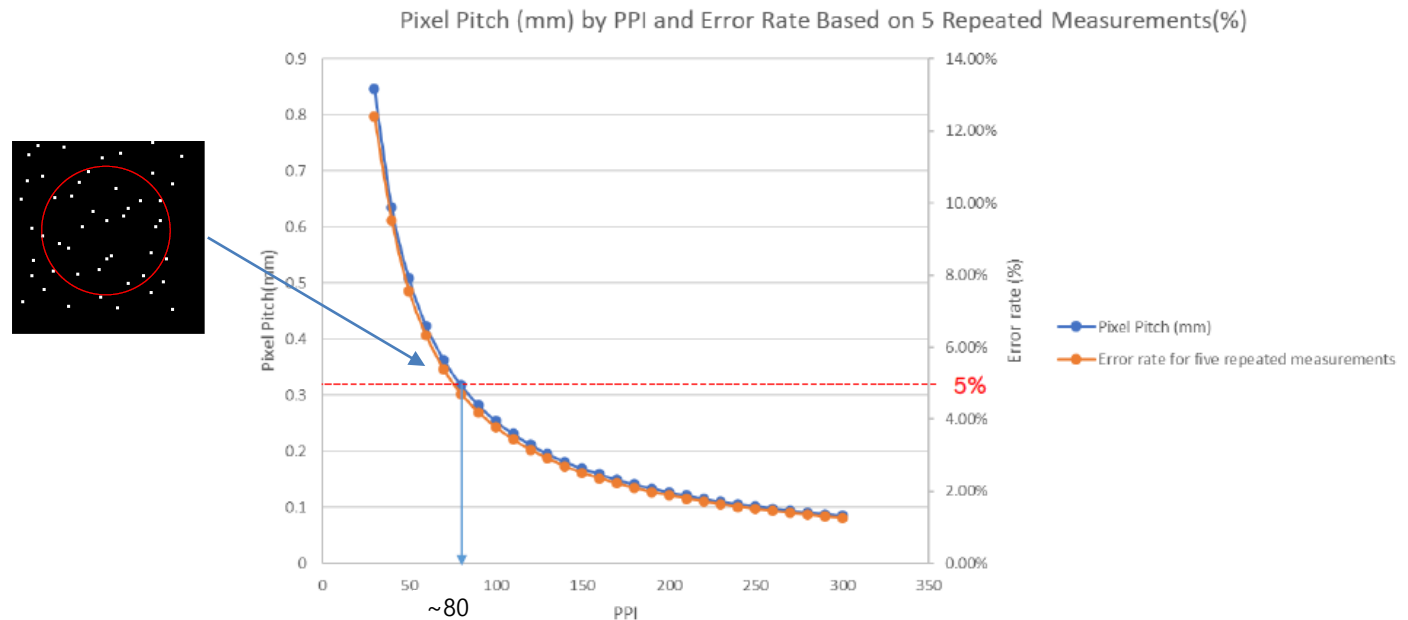
$$Halo_{stops} = \log_2 \left(\frac{L_1}{L_{h2} - L_{h3}} \right), L_{h2} \neq L_{h3}$$



Starfield

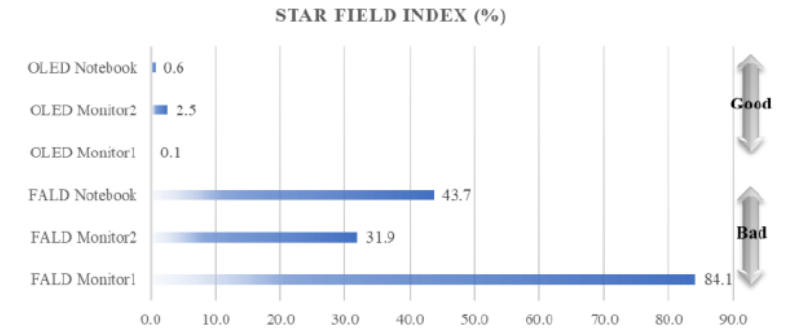
❑ Fixed measurement area: 27mm

- Ensures sufficient sample of stars with 5% error rate, ~80 PPI and above



Starfield

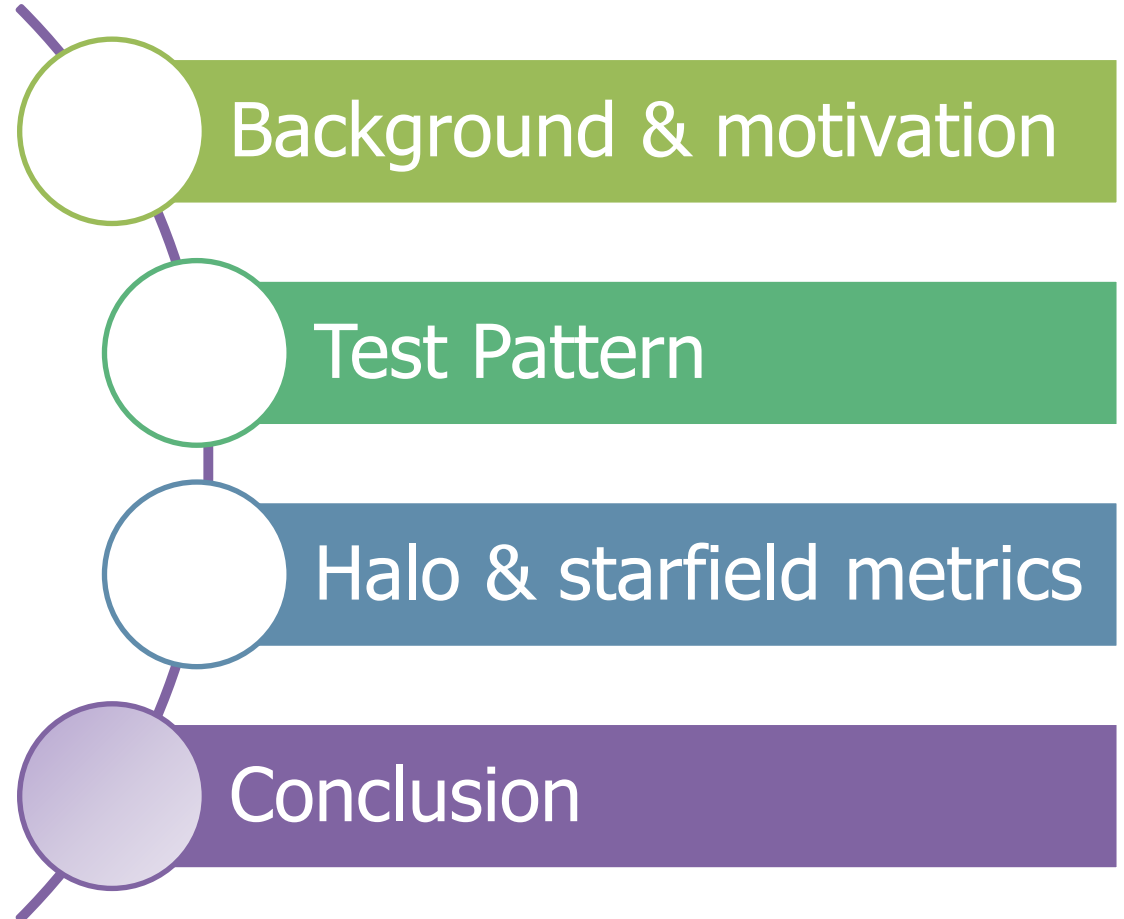
- ❑ L_{sf2} & L_{sf3} are averaged over 5 different random seed
- ❑ L_{id} is ideal value, divided by the pixel density of the starfield
- ❑ High SFI indicates more severe artifacts (worse); lower SFI is better



$$SFI \text{ (Star Field Index, \%)} = \frac{|L_{sf2} - L_{sf3}|}{L_{id}} \times 100, L_{id} = \frac{L_1}{64}$$

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Conclusion

❑ Introduce two metrics for Halo and Starfield measurement

- Halo (stops): Higher is better
- SFI (Starfield Index, %): Lower is better

❑ Efficient to measurement

- Based on 3 common test pattern
- Leverage from VESA DisplayHDR
- Compatible with spot-LMD, contact/non-contact
- Accommodate various display size, resolutions, pixel density, etc.
- Independent of backlight configuration and dimming algorithm

